

DEEPAK PATKAR

PhD, Theoretical and Computational Chemistry

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Experience	Post-doctoral Fellow Fujian Institute of Research on the Structure of Matter, Chinese Academy of Sciences, Fuzhou China Research Topics: Copper-Peptide mediated Liquid-Liquid Phase Separation ML-Predictive Design of MOFs for SO ₂ , H ₂ S, and NH ₃ Capture Skills: GROMACS AmberTools LAMMPS RASPA VASP Materials Studio	2025-Present
	Senior Associate at Innodata India Private Limited Role: Subject Matter Expert Responsibilities: Image-based LLM training data curation Skills: Data Curation Data Annotation Quality Assurance Image Analysis Content Review	2024-2025
	Teaching Assistant at Dr. Harisingh Gour Vishwavidyalaya, Sagar, India Responsibilities: Conducted undergraduate chemistry laboratory classes Developed computational chemistry lab assignments Guided students in scientific computing, data analysis, and interpretation of results. Skills: Teaching Mentoring Laboratory Instruction	2021-2023
Academics	Doctor of Philosophy, Chemistry Dr. Harisingh Gour Vishwavidyalaya, Sagar, MP-470003, India Thesis Supervisor: Dr. Milind Madhusudan Deshmukh [link] Thesis: <i>Quantum Chemical Studies on the Mixed Molecular Clusters: Quantitative Estimates of Hydrogen Bond Strength and Cooperativity.</i> [link] Skills: Gaussian AIMAll DAmQT Multiwfn ORCA GAMESS NWChem VMD Avogadro Data Analysis ML Techniques Cheminformatics MTA MP2 Catalysis NCI MESP QTAIM DFT Theoretical Chemistry	2019-2024
	Master of Science, Chemistry Dr. Harisingh Gour Vishwavidyalaya, Sagar, MP-470003, India Project Title: Application of HPLC and Gas Chromatography for Chemical Analysis. Project Supervisor: Dr. Abhilasha Durgavanshi [link] Skills: Linux Python FORTRAN Bash Scripting HPLC GC-MS Analytical Chemistry Natural Product Chemistry Quantum Chemistry Thermodynamics Spectroscopy Material Chemistry	2017-2019
	Bachelor of Science, Triple major in Physics, Chemistry, Criminology and Forensic Science Dr. Harisingh Gour Vishwavidyalaya, Sagar, MP-470003, India Built foundational understanding of principles, techniques across diverse disciplines. General Classical Physics General Quantum Physics Physical Chemistry Organic Chemistry Inorganic Chemistry Analytical Chemistry Forensic Science Toxicology Criminology Crime Scene Investigation	2014-2017
	Research Interest Theoretical & Computational Chemistry Quantum Chemistry Density Functional Theory Noncovalent Interactions Long-Range Correlation Effects Transition Metal Catalysis Molecular & Biomolecular Simulations Atomistic Machine Learning Inverse Molecular Design Cheminformatics & Bioinformatics Statistical Mechanics Mathematical Physics Energy Storage Materials Quantum Materials & Quantum Computing Applications Machine Learning for Molecular Science CADD	
Programmes	iHBEQuant CuBind [GitHub]	
Publications	Total Publication = 8 , h-index = 6 , i10-index = 5 , Citations = 144 [Google Scholar] - Ismail, T. M.; Patkar, D.; Sajith, P. K.; Deshmukh, M. M. Hydrogen bond strengths in microhydrated clusters of HNO and HONO: Energetic insights via molecular tailoring approach <i>New J. Chem.</i> 2024, 48, 11591–11605. [DOI] - Ismail, T. M.; Patkar, D.; Sajith, P. K.; Deshmukh, M. M. Interplay of Hydrogen, Pnicogen and Chalcogen Bonding in X(H₂O)_n=1-5 (X = NO, NO⁺, and NO⁻) Complexes: Energetics Insights via Molecular Tailoring Approach <i>J. Phys. Chem. A</i> 2023, 127,10360–10374. [DOI] Patkar, D.; Chopra, D.; Deshmukh, M. M. Characterization of non-covalent contacts in mono- and di-halo substituted acetaldehydes: probing the substitution effects of electron donating and withdrawing groups <i>Phys. Chem. Chem. Phys.</i> 2023, 25, 2946. [DOI] Patkar, D.; Ahirwar, M. B.; Deshmukh, M. M. A Tug of War between the Self- and Cross-associating Hydrogen Bonds in Neutral Ammonia-Water Clusters: Energetic Insights by Molecular Tailoring Approach. <i>ChemPhysChem</i> 2022, 23, e202200476. [DOI] Patkar, D.; Ahirwar, M. B.; Deshmukh, M. M. Energetic Ordering of Hydrogen Bond Strengths in Methanol-Water Clusters: Insights via Molecular Tailoring Approach <i>ChemPhysChem</i> 2022, 23, e202200143. [DOI]	

	6. Patkar, D.; Ahirwar, M. B.; Shrivastava, S. P.; Deshmukh, M. M. Assessment of Hydrogen Bond Strengths and Cooperativity in Self- and Cross-Associating Cyclic (HF)_m(H₂O)_n (m + n = 2 to 8) Clusters. <i>New J. Chem.</i> 2022, 46, 2368–2379. [DOI] 7. Patkar, D.; Ahirwar, M. B.; Gadre, S. R.; Deshmukh, M. M. Unusually Large Hydrogen-Bond Cooperativity in Hydrogen Fluoride Clusters, (HF)_n, n = 3 to 8, Revealed by the Molecular Tailoring Approach. <i>J. Phys. Chem. A</i> 2021, 125, 8836–8845. [DOI] 8. Ahirwar, M. B.; Patkar, D.; Yadav, I.; Deshmukh, M. M. Appraisal of Individual Hydrogen Bond Strengths and Cooperativity in Ammonia Clusters via Molecular Tailoring Approach. <i>Phys. Chem. Chem. Phys.</i> 2021, 23, 17224–17231. [DOI] 9. Patkar, D. Uncovering the Individual Hydrogen Bond Strengths and Cooperativity in Pure (H₂S)_n and Mixed (H₂O)_m(H₂S)_n (m + n = 2 to 8) Clusters. <i>ChemRxiv</i> 2025. [DOI] [Pre-print] 10. Patkar, D. From Weak to Functional: Network Cooperativity and Anti-Cooperativity in Cyclic Neutral (HF)_m(NH₃)_n Hydrogen Bonded Clusters, <i>J. Phys. Chem. A</i> 2026, xxx,xxxx–xxxx. [Submitted]
Selected Talks	1. National Seminar on “Recent Advances in Chemical Sciences”, Department of Chemistry, Dr. Harisingh Gour Vishwavidyalaya Sagar, March, 2023. [link] Title: Competition between Self- and Cross-associating Hydrogen Bonds in Neutral Ammonia-Water Clusters: Energetic Insights by Molecular Tailoring Approach. [link] 2. DAE Symposium on “Current Trends in Theoretical Chemistry, Chemistry Division, Bhabha Atomic Research Centre BARC, Mumbai-400085, September, 2022. [link] Title: Energetic Ordering of Hydrogen Bond Strengths in Methanol-Water Clusters: Insights via Molecular Tailoring Approach. [link] 3. DAE Symposium on “Current Trends in Theoretical Chemistry, Chemistry Division, Bhabha Atomic Research Centre BARC, Mumbai-400085, September, 2021. [link] Title: Evaluation of individual H-Bond Strengths and Cooperativity in (HF)_m(H₂O)_n, (m + n) = 2 to 8 Clusters. [link] 4. Theoretical Chemistry Symposium, IISER-Kolkata, IACS Kolkata, December, 2021. [link] Title: Estimation of Individual Hydrogen Bond Strengths and Cooperativity in (HF)_m(H₂O)_n, (m + n) = 2 to 8 Clusters via Molecular Tailoring Approach. [link] 5. International Seminar, Dr. Harisingh Gour Vishwavidyalaya Sagar, January, 2021. Title: Theoretical Study of Hydrosilylation of Olefin Catalyzed by Cationic Calcium Hydride Catalyst. [link]
Selected Conferences	1. National Seminar on “Recent Advances in Chemical Sciences”, Department of Chemistry, Dr. Harisingh Gour Vishwavidyalaya Sagar, March 2023. [link] 2. DAE Symposium on “Current Trends in Theoretical Chemistry, Chemistry Division, Bhabha Atomic Research Centre BARC, Mumbai-400085, September 2022. [link] 3. Hand on Workshop “Molecular Tailoring Approach”, Department of Scientific Computing, Modelling and Simulation, Savitribai Phule Pune University, May 2022. [link] 4. Theoretical Chemistry Symposium, IISER-Kolkata, IACS Kolkata, Kalyani University, December, 2021. [link] 5. DAE Symposium on “Current Trends in Theoretical Chemistry, Chemistry Division, Bhabha Atomic Research Centre BARC, Mumbai-400085, September 2021. [link]
Other Skills	Quantum Chemistry Molecular Modeling Molecular Dynamics Simulations Scientific Programming Python Bash FORTRAN Linux High-Performance Computing Machine Learning Atomistic Machine Learning Cheminformatics Bioinformatics Data Analytics Quantum Computing Q-Chem PySCF PyMatGen RDKit Discovery Studio CP2K PyMOL ChimeraX VESTA OVITO ChemCraft Spartan GaussView ChemDraw LaTeX Gnuplot OriginPro
Achievements	INSPIRE Scholarship, DST India (2014) Best Oral Presentation Award, DHSGSU (2023)
Language	English Hindi Sanskrit
References	1. Dr. Milind Deshmukh Ph.D. Supervisor Associate Professor, Department of Chemistry, University of Delhi, Delhi Mobile: +91 8989180813, Email: milind.deshmukh@gmail.com webpage: [link] 2. Prof. Deepak Chopra Crystallography and Crystal Chemistry Laboratory, Department of Chemistry, IISER Bhopal Office: +91-755-669-1311; Fax: +91 755 6692392, Email: dchopra@iiserb.ac.in webpage: [link] 3. Dr. Pookkottu K. Sajith Assistant Professor, Department of Chemistry, Farook College, Kozhikode, Kerala Mobile: +91 9349280803, Email: pkajith@farookcollege.ac.in webpage: [link]